

LOAN APPROVAL PREDICTION SYSTEM

Machine Learning for Automated Credit Risk Assessment

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Objective

1

Automate Decisions

Reduce reliance on manual, subjective loan underwriting.

2

End-to-End Pipeline

Ingest, clean, engineer features, train, evaluate, deploy.

3

Compare Models

Benchmark multiple classifiers, select the best performer.

4

Real-Time Interface

Serve predictions instantly via a Streamlit web app.

Methodology



Supervised binary classification: each applicant is an independent observation with input features and a known outcome label. The model learns the statistical relationship between attributes (income, credit history, loan amount) and the approval decision, generalizing to new applicants.

Dataset

614

Total Records

12

Features

80 / 20

Train / Test Split

CSV

Data Format

Source: Loan Prediction Problem Dataset (Kaggle)

Feature	Description
Gender, Married, Dependents	Applicant demographics
Education, Self_Employed	Employment / education status
ApplicantIncome, CoapplicantIncome	Monthly income figures
LoanAmount, Loan_Amount_Term	Requested amount and repayment term
Credit_History, Property_Area	Credit standing and location
Loan_Status	Target label: Approved (Y) / Rejected (N)

Algorithm — Process Flow



Models Compared

Logistic Regression

Interpretable linear baseline

Random Forest

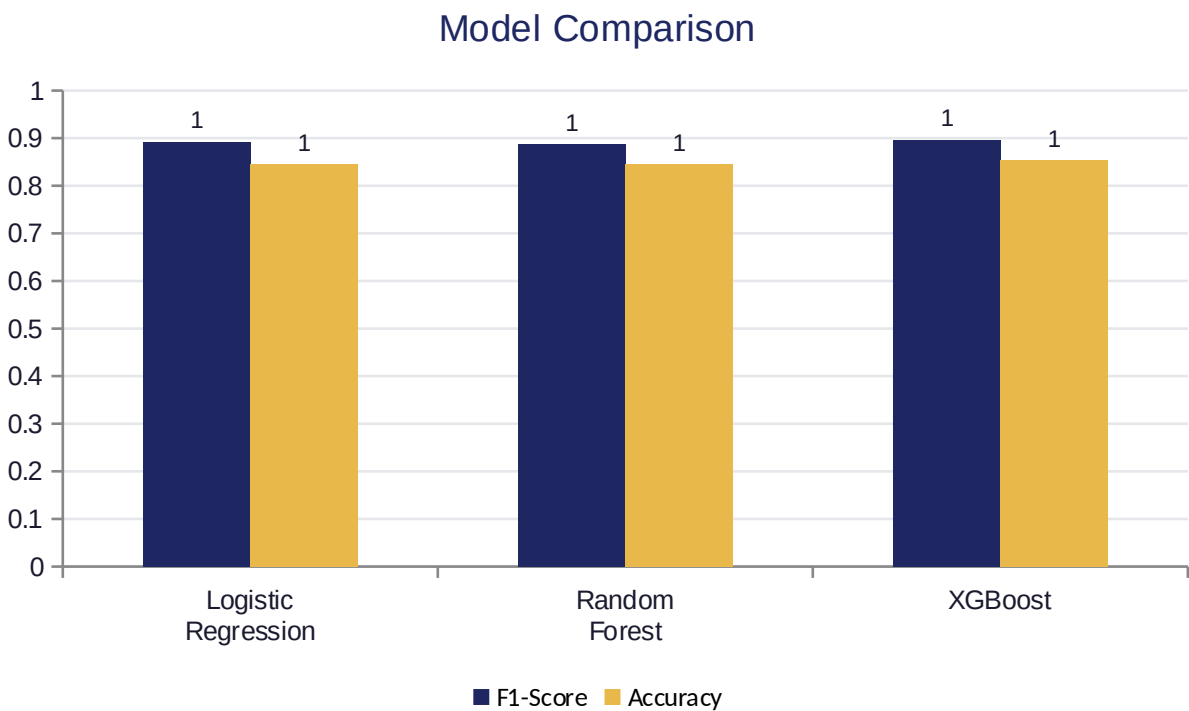
Ensemble of trees; captures non-linear interactions

XGBoost

Gradient boosting; corrects prior errors sequentially

Hyperparameters tuned via GridSearchCV with 5-fold stratified cross-validation; SMOTE applied to correct class imbalance before training.

Results



Best Model: XGBoost



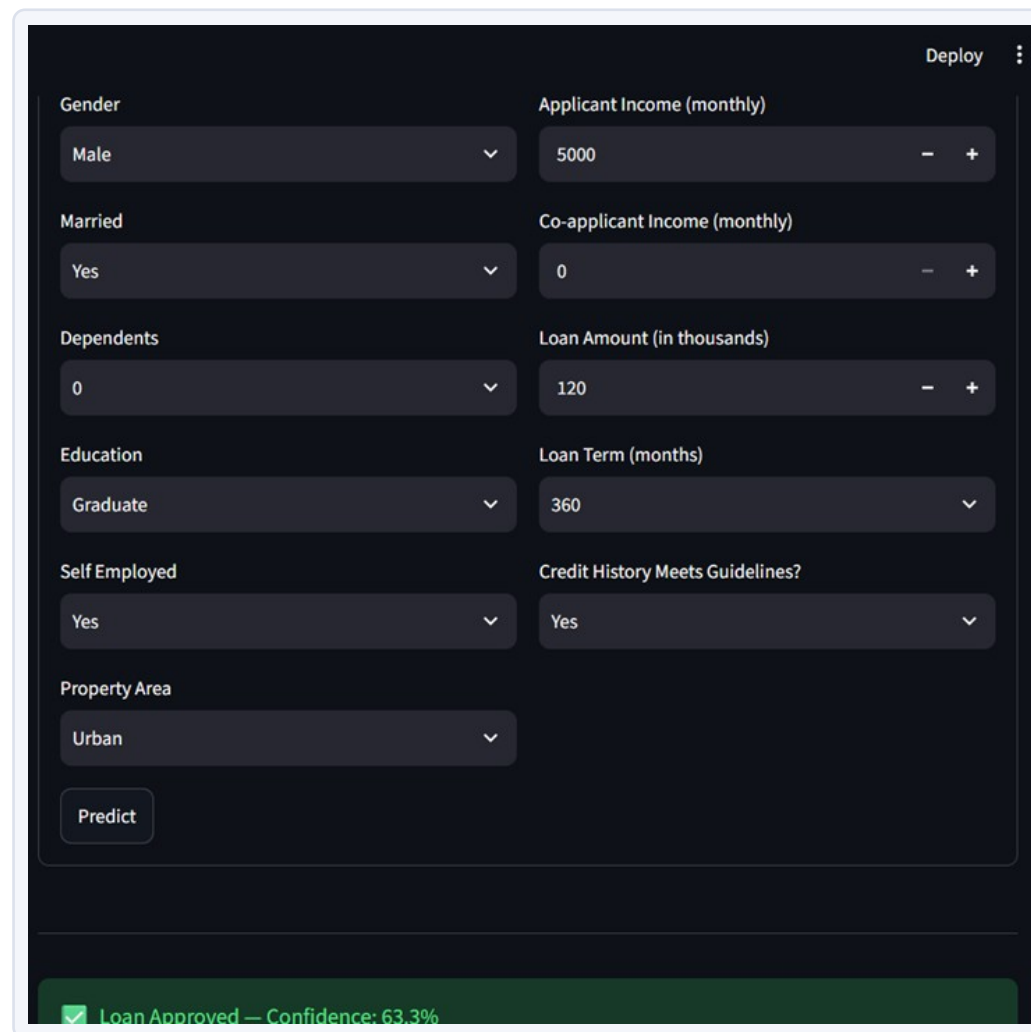
Test Set Breakdown (123 records)

Class	Precision	Recall	F1
Rejected	0.79	0.71	0.75
Approved	0.88	0.92	0.90

XGBoost was selected as the final model — highest F1-score, balancing precision and recall across both classes — and its test accuracy (85.37%) falls within the project's 80–90% target range.

Live Prediction Demo

Deployed Streamlit interface — real-time loan approval prediction with confidence score.



The image shows a Streamlit web application interface for loan approval prediction. The interface is dark-themed and contains several input fields for user data. At the top right, there is a 'Deploy' button and a menu icon. The inputs are arranged in two columns:

- Gender:** A dropdown menu with 'Male' selected.
- Applicant Income (monthly):** A numeric input field with '5000' and minus/plus buttons.
- Married:** A dropdown menu with 'Yes' selected.
- Co-applicant Income (monthly):** A numeric input field with '0' and minus/plus buttons.
- Dependents:** A dropdown menu with '0' selected.
- Loan Amount (in thousands):** A numeric input field with '120' and minus/plus buttons.
- Education:** A dropdown menu with 'Graduate' selected.
- Loan Term (months):** A dropdown menu with '360' selected.
- Self Employed:** A dropdown menu with 'Yes' selected.
- Credit History Meets Guidelines?:** A dropdown menu with 'Yes' selected.
- Property Area:** A dropdown menu with 'Urban' selected.

At the bottom left, there is a 'Predict' button. At the bottom of the interface, a green status bar displays the result: '✅ Loan Approved — Confidence: 63.3%'.

Conclusion

- Complete, working ML pipeline — from raw Kaggle data to a deployed prediction interface.
- XGBoost selected as the final model with 85.37% test accuracy, within the project's target range.
- Balanced performance across both approved and rejected classes, validated via cross-validation.
- Real-world potential: faster loan processing, more consistent underwriting, fairer lending decisions.
- Future work: explainability (SHAP/LIME) for transparent, individual-level prediction reasoning.